

Student Exercise: Improve the Fibonacci Function

Background

The following function calculates Fibonacci numbers using recursion. It works, but it calculates some Fibonacci values many times.

```
int Fib(int n)
{
    if (n == 0)
    {
        return 0;
    }
    else if (n == 1)
    {
        return 1;
    }
    else
    {
        return Fib(n - 1) + Fib(n - 2);
    }
}
```

Your Task

1. Rewrite the Fib function so that it does not use recursion.
2. Use a for loop to calculate the Fibonacci number.
3. Make sure each Fibonacci number is calculated only once.
4. The function must still return the correct Fibonacci number.

Test your function with this code:

```
for (int i = 0; i < 10; i++)
{
    Console.WriteLine(i + ": " + Fib(i));
}
```

Expected output:

```
0: 0
1: 1
2: 1
3: 2
4: 3
5: 5
6: 8
7: 13
8: 21
9: 34
```

Hints

- You do not need an array.
- Keep track of only the previous two Fibonacci numbers.
- Start with: first = 0 and second = 1.
- Calculate the next value using: $\text{next} = \text{first} + \text{second}$.
- After calculating next, move the values forward and repeat using a for loop.

Think about this sequence:

first	second	next
0	1	1
1	1	2
1	2	3
2	3	5